Frederick house sales 2019

Daniel Coblentz

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Packages

In addition to our usual `tidyverse` package, we load

- `scales` for making nice axis labels with variables measured in dollars. Use `+ scale_x_continuous(labels = label_dollar())`.
- `broom` for using models to "augment" data sets with fitted values and residuals. For details, see the demo.

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor
```

```
library(broom)
```

The data

We will study a dataset that includes records of sales of residential properties in Frederick County from 2019.

The next chunk loads the data and adjusts two of the variables to make them a bit easier to graph and summarize.

```r
frederick_2019 <- read_csv("data/frederick_2019.csv") %>%
  mutate(basement = factor(basement, levels = c("NO", "YES", "FINISH")),
         heat = fct_infreq(heat))
```

```
## Rows: 3284 Columns: 9
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (2): basement, heat
## dbl (6): price, land_area, building_area, full_baths, half_baths, condition
## lgl (1): has_fireplace
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Brief codebook

The cases in the data set are the sales of residential properties in Frederick County in 2019. If a property was sold multiple times in 2019, only the last sale is recorded. A relatively small number of sales (on the order of 100) were removed from the dataset since their records were incomplete.

Variables

- `price`: the sales price of the property, in USD.
- `land_area`: the area of the land associated with the property, measured in acres
- `building_area`: the area of the main building on the property, not including "below grade" space such as a basement, in square feet.
- `heat`: the type of heat for the main building
- `full_bath`: the number of full baths (bathrooms with bathing facilities) in the main building
- `half_bath`: the number of half baths (bathrooms without bathing facilities) in the main building
- `condition`: condition of the main building on an 8-point scale, where 8 is "excellent"
- `has_fireplace`: whether the main building has at least one fireplace

0. Examine the dataset in the usual ways and familiarize yourself with the variables and their types.

```r
head(frederick_2019, 5)
```

```
## # A tibble: 5 x 9
##    price land_area building_area basement heat   full_baths half_baths condition
##    <dbl>     <dbl>         <dbl> <fct>    <fct>       <dbl>      <dbl>     <dbl>
## 1  50000      0.74          1600 YES      SPACE           1          0         3
## 2  75000      0.25           992 YES      SPACE           0          0         2
## 3 300000      0.43          1080 FINISH   ELEC            2          1         4
## 4  40000      0.12           880 NO       SPACE           1          0         2
## 5 332000      0.22          2080 YES      HT PMP          2          1         4
## # i 1 more variable: has_fireplace <lgl>
```

```r
glimpse(frederick_2019)
```

```
## Rows: 3,284
## Columns: 9
## $ price         <dbl> 50000, 75000, 300000, 40000, 332000, 318500, 265000, 425~
## $ land_area     <dbl> 0.7400, 0.2500, 0.4300, 0.1200, 0.2200, 2.4200, 1.3900, ~
## $ building_area <dbl> 1600, 992, 1080, 880, 2080, 1298, 1092, 2136, 1196, 1804~
## $ basement      <fct> YES, YES, FINISH, NO, YES, YES, YES, YES, YES, YES, YES,~
```

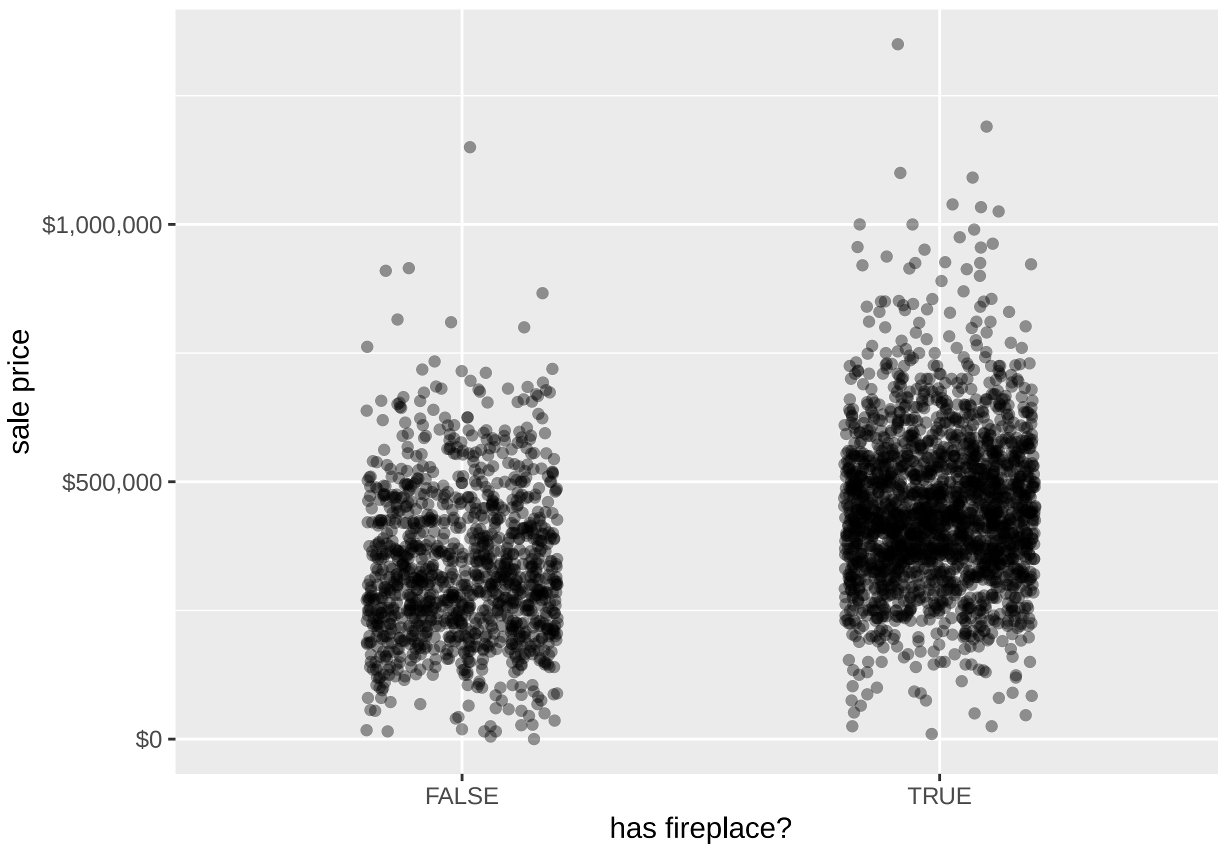
```
## $ heat          <fct> SPACE, SPACE, ELEC, SPACE, HT PMP, HT PMP, HT PMP, HT PM~
## $ full_baths    <dbl> 1, 0, 2, 1, 2, 2, 2, 3, 1, 2, 2, 1, 1, 2, 1, 1, 2, 3, 1,~
## $ half_baths    <dbl> 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 1,~
## $ condition     <dbl> 3, 2, 4, 2, 4, 4, 4, 4, 3, 4, 4, 3, 4, 3, 4, 4, 4, 5, 5,~
## $ has_fireplace <lgl> TRUE, FALSE, TRUE, FALSE, FALSE, TRUE, TRUE, TRUE, FALSE~
```

Here, we are interested in the sales price of the houses again, this time, as predicted by whether the house has a fireplace or not.

1. Classify `price` and `has_fireplace` according to whether they are categorical or numerical.

'price' is numerical, and 'has_fireplace' is categorical ### 2. Make a scatterplot of `price` versus the `has_fireplace` variable, taking steps to address overplotting.

```
ggplot(frederick_2019, aes(x = has_fireplace, y = price)) +
  geom_jitter(width = 0.2, alpha = 0.4) +
  scale_y_continuous(labels = label_dollar()) +
  labs(x = "has fireplace?", y = "sale price")
```



3. Calculate some summary statistics for `price`, grouped by `has_fireplace`. Include the mean, median, standard deviation, and the first and third quartiles.

```
frederick_2019 %>%
  group_by(has_fireplace) %>%
  summarise(
    mean_price   = mean(price),
    median_price = median(price),
    sd_price     = sd(price),
```

```r
    q1_price     = quantile(price, 0.25),
    q3_price     = quantile(price, 0.75)
  )
```

```
## # A tibble: 2 x 6
##   has_fireplace mean_price median_price sd_price q1_price q3_price
##   <lgl>              <dbl>        <dbl>    <dbl>    <dbl>    <dbl>
## 1 FALSE            331483.       310000  146790.   225000  422854
## 2 TRUE             445662.       434900  150022.   350000  530164.
```

4. Write a description of what you see in the graph, comparing the sales prices of houses with and without fireplaces. Support what you say with the summary statistics you have calculated.

The graph shows homes that have fireplaces and ones that do not against the sales price. We can see generally homes with a fireplace included apppear to have higher selling prices that ones without. This can also be verified by both the mean and median prices from the table above.

5. Use `lm` to make a model of price using your categorical variable as predictor. Print the output for the model.

```r
model_fireplace <- lm(price ~ has_fireplace, data = frederick_2019)
summary(model_fireplace)
```

```
##
## Call:
## lm(formula = price ~ has_fireplace, data = frederick_2019)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -435662  -96662  -15662   88485  904338
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)         331483       4354   76.13   <2e-16 ***
## has_fireplaceTRUE   114179       5426   21.04   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 148900 on 3282 degrees of freedom
## Multiple R-squared:  0.1189, Adjusted R-squared:  0.1186
## F-statistic: 442.8 on 1 and 3282 DF,  p-value: < 2.2e-16
```

6. Write the equation for the model using the model summary output.

price_hat = 331483 + 114179x

where x is binary (1 or 0) for if the variable has_fireplace is true or false.

7. Interpret the "intercept" and "slope" for this model. In particular, see if you can figure out what the relationship is between the slope, intercept, and the statistics you calculated earlier. We will discuss this as a class after you try.

The intercept represents the predicted average sale price of houses without a fireplace it is the baseline price when the variable has_fireplace = false Additionally, the slope represents the estimated increase in sales price for homes with a fireplace. The slope also corresponds to the difference in mean house prices between homes with and without a fireplace. This corresponds with our summary statistics above, where the mean price for homes with a fireplace was higher than for those without.